
TECHNICAL REPORT

HYBRID-ELECTRIC PROPULSION & MISSION ENDURANCE OPTIMIZATION

AeroOptima Sizing and Simulation Platform for a 1000 kg Fixed-Wing Tactical UAV

A genetic-algorithm-driven design tool coupling a physics-based flight-mission simulator with an evolutionary search over engine, battery, and motor sizing — with a reinforcement-learning power-management policy evaluated against a rule-based baseline.

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1. Executive Summary

AeroOptima is a design and simulation platform for sizing the hybrid-electric propulsion system of a fixed-wing tactical unmanned aerial vehicle (UAV) with a maximum takeoff weight (MTOW) of 1,000 kg. The platform's central engineering question is: *for a given mission profile and operating environment, what turboshaft engine rating, battery pack capacity, and motor count maximize total flight endurance while remaining within the weight budget and airworthiness constraints of the airframe?*

The system answers this question by coupling two loops. An **outer optimization loop** implemented as a genetic algorithm (GA) using the DEAP framework searches the three-dimensional design space of engine power, battery capacity, and motor count (with an optional extension to also search the cruise/loiter power-split ratio). Each candidate design is scored by an **inner simulation loop** — a custom physics environment built on the Gymnasium interface that flies the candidate aircraft through a complete, user-defined multi-leg mission (climb, cruise, loiter, descent, landing) under a full non-linear aerodynamic, propulsive, and atmospheric model, including compressibility effects, engine altitude derating, battery thermal derating, and a stall-limited flight envelope. The GA's fitness signal is the resulting mission endurance in hours, penalized if the aircraft fails to land safely or fails to meet an on-station requirement.

A rule-based energy management strategy (EMS) governs the instantaneous split of power between the turboshaft engine and the battery-electric motor across mission phases, prioritizing electric power for low-speed/high-torque phases and engine power for sustained cruise. A reinforcement-learning (PPO) policy trained on the same environment under randomized environmental disturbances is provided as an alternative EMS and benchmarked against the heuristic under scripted storm/cold/turbulence conditions.

The platform is delivered as a FastAPI backend exposing a single mission-optimization endpoint and a Next.js/React dashboard providing mission authoring, a 3D flight-path replay, and telemetry visualization. The implementation is validated by a deterministic regression suite pinned against a golden-baseline result set, by unit-level physics sanity checks, and by an independent multi-scenario stress test spanning six distinct operating theatres (high-altitude cold, hot desert, temperate plains, monsoon-belt turbulence, and maritime patrol). This report documents the system's architecture, physical modelling assumptions, optimization methodology, implementation, and validation results, and states plainly where the current model's assumptions do not yet hold under the most demanding conditions tested.

2. Introduction

2.1 Problem Statement

Hybrid-electric propulsion — pairing a fuel-burning turboshaft engine with a battery-electric motor driving a common propeller — is an attractive architecture for tactical UAVs because it combines the high energy density of liquid fuel (long-range cruise) with the low acoustic signature and instant-torque response of electric propulsion (stealthy loiter, robust takeoff/climb margin). The central engineering trade-off is sizing: an oversized engine and battery waste weight that could otherwise be fuel or payload, while an undersized combination fails to meet the mission's power or endurance requirements. Because the "best" size depends on the specific mission profile (altitude, speed, distance, loiter duration) and environment (temperature, wind, turbulence), a fixed one-size-fits-all propulsion package is inherently sub-optimal across the range of missions a tactical UAV may be tasked with.

AeroOptima addresses this by making sizing itself the output of an optimization process driven by an explicit, physically-grounded mission and environment specification, rather than a static reference design.

2.2 Design Requirements and Constraints

Maximum takeoff weight (MTOW)	1000 kg (hard constraint)	Airframe (structural) mass	350 kg (fixed)
Payload capacity	50 – 350 kg (configurable, default 200 kg)	Wing area / wingspan	14.0 m² / 15.0 m
Propulsion architecture	Parallel hybrid – turboshaft + electric motor on a shared propeller	Engine sizing range (GA-searched)	30 – 120 kW
Battery sizing range (GA-searched)	16.69 – 50.0 kWh	Motor count (GA-searched)	1, 2, or 4
Battery-only reserve requirement	STANAG 4671-style 30-minute single-fault reserve		

The lower bound of the battery search space (16.69 kWh) is not arbitrary: it is derived from the 30-minute battery-only reserve requirement at an average loiter draw of 27.24 kW, over the pack's usable state-of-charge (SoC) window of 10%–95%. This ties one boundary of the search space directly to an airworthiness requirement rather than a component catalogue limit.

3. System Architecture

3.1 Technology Stack

Layer	Technology	Role
Simulation core	Python 3.10+, Gymnasium, NumPy, SciPy	Physics-based flight/energy simulation exposed as an RL-style environment
Optimization	DEAP (Distributed Evolutionary Algorithms in Python)	Genetic algorithm sizing loop
Learned policy (optional)	PyTorch, Stable-Baselines3 (PPO)	Reinforcement-learning energy-management policy, lazily imported
Backend API	FastAPI, Pydantic v2, Uvicorn	REST endpoint, request validation, response schemas
Frontend	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS 4	Mission authoring dashboard and results visualization
Visualization	Three.js (react-three-fiber/drei), Plotly.js	3D flight-path replay and telemetry charting
Testing	pytest	Physics smoke tests and golden-baseline regression suite
Packaging	uv / pyproject.toml (backend), npm (frontend)	Dependency management

3.2 Repository Organization

```
HALxOPT_D/  
├─ backend/  
│   ├── environment.py    - UAVHybridEnv: physics core (Gymnasium environment)  
│   ├── optimizer.py      - Genetic algorithm sizing loop (DEAP)  
│   ├── main.py           - FastAPI REST server / mission re-simulator  
│   ├── sweep.py          - ±10% SFC / battery-density sensitivity sweep  
│   ├── rl_policy.py      - RL feature extraction + PPO inference wrapper  
│   └─ rl/  
│       ├── train_policy.py - PPO training on randomized disturbance scenarios  
│       └─ benchmark_policy.py - heuristic-vs-RL disturbance benchmark  
│   ├── test_env.py       - physics smoke tests  
│   └─ test_regression.py - golden-baseline regression tests  
├─ data/  
│   ├── aerodynamics.json, turboshaft_specs.json,  
│   └─ battery_specs.json, motor_specs.json    - component specification database
```

```
└─ frontend/
  └─ src/
    └─ app/                                - Next.js App Router entry
      └─ components/
        ├── Dashboard.tsx                 - state management, mission builder, results panels
        ├── FlightScene.tsx               - 3D flight-path visualization (Three.js)
        ├── TelemetryChart.tsx            - Power/Resources/Trajectory charts (Plotly)
        └─ TelemetryTable.tsx             - per-timestep telemetry matrix
```

3.3 Data-Driven Configuration Model

All physical component parameters — airframe geometry, aerodynamic coefficients, turboshaft ratings and fuel consumption, motor ratings, and battery characteristics — are externalized into four JSON specification files under `data/` and loaded once at simulation-environment construction. No physical constant is hardcoded inside the simulator itself; changing a component's rating or efficiency is a data change, not a code change. This also allows a battery chemistry preset (`Li-NCA` default, `Li-LFP` alternative) to be swapped at request time without altering the search space.

4. Physical and Environmental Model

4.1 Atmosphere Model

The simulator implements an International Standard Atmosphere (ISA) model generalized to accept a configurable sea-level-equivalent temperature T_0 (default 15 °C, the ISA standard day), so that non-standard-day and cold/hot climates can be modelled directly through a single input parameter rather than a separate model.

Temperature at altitude (troposphere, $h < 11,000$ m): Eq. 4.1

$$T(h) = T_0 - 0.0065 \cdot h \quad [^{\circ}\text{C}]$$

Density at altitude (troposphere):

$$\rho(h) = \rho_0 \cdot ((273.15 + T(h)) / (273.15 + T_0)) ^ 4.25588$$

Density above the tropopause (stratosphere, $h \geq 11,000$ m):

$$\rho(h) = \rho(11000) \cdot \exp(-(h - 11000) / 6341.6)$$

Density floor: $\rho \geq 0.05 \text{ kg/m}^3$ (always enforced)

At the ISA standard day ($T_0 = 15$ °C) this reduces algebraically to the classical fixed-temperature ISA density formula, so the generalization introduces no discrepancy for standard-day analysis while allowing cold/hot baselines to be modelled consistently.

Speed of sound and Mach number: Eq. 4.2

$$a(T) = \sqrt{1.4 \cdot 287.05 \cdot (273.15 + T)} \quad [\text{m/s}]$$
$$M = V / a$$

Stall speed:

$$V_{\text{stall}} = \sqrt{2 \cdot W \cdot g / (\rho \cdot S \cdot CL_{\text{max}})} \quad [CL_{\text{max}} = 1.5]$$

4.2 Aerodynamic Model

Lift-induced and profile drag follow a classical Oswald-efficiency drag polar, with an added compressibility correction that engages only in the transonic-approach regime.

$$\gamma = \text{asin}(\text{clip}(\text{climb_rate} / V, -1, 1)) \quad [\text{flight-path angle}] \quad \text{Eq. 4.3}$$
$$CL = 2 \cdot W \cdot g \cdot \cos(\gamma) / (\rho \cdot V^2 \cdot S) \quad [\text{lift coefficient}]$$
$$CD_{\text{induced}} = CL^2 / (\pi \cdot AR \cdot e) \quad [AR = b^2 / S = 16.07]$$

Prandtl-Glauert compressibility correction ($0.4 < M < 0.95$ only):

$$CD_{\text{mach}} = CD_0 \cdot (1 / \sqrt{1 - M^2} - 1), \quad \text{else } 0$$
$$CD = CD_0 + CD_{\text{induced}} + CD_{\text{mach}}$$

```

P_aero = 0.5 · ρ · V³ · S · CD · turbulence_factor    [W]
P_climb = W · g · climb_rate                          [W]
P_shaft = (P_aero + P_climb) / η_prop                 [W, floored at 0]

```

Turbulence is modelled as a multiplicative penalty on aerodynamic power draw, `turbulence_factor` = `clip(1 + N(0, 0.05·level), 0.8, 1.2)`, sampled per timestep from the configured turbulence level (0 = calm, 1 = rough), so its effect is stochastic rather than a fixed derate.

4.3 Propeller Efficiency Model

Propeller efficiency is computed dynamically from an advance-ratio model rather than a fixed per-phase constant, and falls back to a static lookup table only at near-zero airspeed (ground roll / hover-like conditions) where the advance-ratio model is not meaningful.

```

n = RPM / 60                                [RPM fixed at 2200]
J = V / (n · D)                             [D = propeller diameter = 1.8 m]
η = 4·η_max·(J/J_opt)·(1 - J/(2·J_opt))      [η_max = 0.86, J_opt = 0.85]
η_prop = clip(η, 0.50, η_max)

```

Eq. 4.4

Static fallback ($V < 1$ m/s): 0.65 (takeoff) / 0.75 (climb) / 0.85 (cruise, loiter) / 0.70 (descent, landing)

5. Propulsion Subsystem Models

5.1 Turboshaft Engine

The engine model scales continuously with the GA-searched design variable `engine_size_kw` against a fixed reference unit (67.5 kW / 35 kg), so both weight and rated power scale proportionally with the sized engine rather than being capped at the reference unit's flat rating:

$$\begin{aligned} \text{weight_engine} &= (\text{engine_size_kw} / 67.5) \cdot 35 \\ \text{engine_continuous_kw} &= 0.8889 \cdot \text{engine_size_kw} \quad [= \min(\text{engine_size_kw}, 60 \cdot (\text{engine_size_kw}/67.5))] \\ \text{engine_peak_kw} &= 1.1111 \cdot \text{engine_size_kw} \quad [= \min(\text{engine_size_kw} \cdot 1.25, 75 \cdot (\text{engine_size_kw}/67.5))] \end{aligned} \quad \text{Eq. 5.1}$$

Available engine power is derated with altitude using the **Gagg–Ferrar** model (a classical 1934 empirical relation for naturally-aspirated engine power loss with air density), applied separately to the continuous and peak ratings, with a floor preventing the engine from being modelled as fully dead purely from altitude:

$$\begin{aligned} \sigma &= \rho(h) / \rho_0 \\ \text{derate} &= \text{clip}(\sigma - (1 - \sigma)/7.55, 0.15, 1.0) \\ P_{\text{max_engine}}(h) &= P_{\text{rating}} \cdot \text{derate} \end{aligned} \quad \text{Eq. 5.2}$$

Fuel consumption follows a base specific fuel consumption (SFC) of 0.38 kg/kWh (Jet-A1), with a partial-load penalty that reflects a real turboshaft's efficiency loss when operated well below its rated continuous power:

$$\begin{aligned} \text{load_fraction} &= P_{\text{engine}} / \text{engine_continuous_kw} \\ \text{load_fraction} \geq 0.8: & \quad \text{SFC} = \text{SFC_base} \quad (\text{optimal}) \\ 0.5 \leq \text{load_fraction} < 0.8: & \quad \text{SFC} = \text{SFC_base} \cdot (1 + 0.15 \cdot (0.8 - \text{load_fraction})/0.3) \quad (\text{up to } +15\%) \\ \text{load_fraction} < 0.5: & \quad \text{SFC} = \text{SFC_base} \cdot (1.15 + 0.25 \cdot (0.5 - \text{load_fraction})/0.5) \quad (\text{up to } +40\%) \\ \text{fuel_burned_kg} &= \text{SFC_eff}(P_{\text{engine}}) \cdot P_{\text{engine}} \cdot \text{dt_hours} \end{aligned} \quad \text{Eq. 5.3}$$

5.2 Electric Motor

The motor reference unit is an EMRAX 228 Medium Voltage machine (12.3 kg, 55 kW continuous / 109 kW peak, 96% peak efficiency, 8.86 kW/kg power density). Motor count is a GA-searched variable (1, 2, or 4 units), and both weight and available electrical-to-shaft power scale linearly with count. The 96% efficiency figure converts electrical draw to shaft power on discharge, and shaft power to battery-charging power (in combination with the generator-efficiency term) during regenerative descent.

5.3 Battery Pack

The reference cell is a 21700-format Li-NCA cylindrical cell (250 Wh/kg, continuous/peak C-rates of 3.0/5.0, usable SoC window 10%–95%, 92% charging efficiency, 85% generator efficiency for engine-driven charging). An alternative Li-LFP preset (160 Wh/kg, 2.5C/4.0C) trades energy density for a chemistry generally considered more cold-tolerant, selectable independently of the GA search.

$$\text{weight_battery} = (\text{battery_capacity_kwh} \cdot 1000) / \text{energy_density_wh_per_kg} \quad \text{Eq. 5.4}$$

Battery power limit (C-rate + motor rating, thermally derated):

$$P_{\text{max_battery}} = \min(\text{battery_capacity_kwh} \cdot \text{temp_penalty} \cdot c_{\text{rate}}, \text{motor_rated_kw})$$

Cold-battery derating (below +10 °C only):

$$T \geq 10^{\circ}\text{C}: \text{penalty} = 1.0$$

$$T < 10^{\circ}\text{C}: \text{penalty} = \text{clip}(1.0 - (10 - T) \cdot 0.01, 0.65, 1.0) \quad [\text{max 35\% loss, at } -25^{\circ}\text{C}]$$

The cold-derating penalty is applied to both usable capacity and instantaneous power limit. No equivalent high-temperature derating currently exists in the model; this is discussed as a known limitation in Section 11.

5.4 Weight Budget and MTOW Closure

The aircraft's total weight is a closed budget against the 1,000 kg MTOW: airframe (fixed 350 kg) plus payload (configurable) plus the sized engine, motor(s), and battery, with the remainder allocated to fuel at a configurable initial fuel fraction. If the sized combination leaves zero or negative fuel budget, the candidate is infeasible and is assigned a fitness of zero by the optimizer without further simulation (Section 8.3).

$$\begin{aligned} \text{weight_empty_and_payload} &= \text{airframe}(350) + \text{payload} + \text{weight_engine} + \text{weight_motor} + \text{weight_battery} \\ \text{fuel_initial} &= (\text{MTOW} - \text{weight_empty_and_payload}) \cdot \text{initial_fuel_fraction} \quad [\text{MTOW} = 1000 \text{ kg}] \end{aligned} \quad \text{Eq. 5.5}$$

6. Mission Simulation Engine

6.1 Environment Formulation

The simulation core, `UAVHybridEnv`, is implemented against the Gymnasium interface — not to train a control policy against a reward signal for the sizing loop itself (the GA uses raw endurance hours as fitness, not the RL reward), but because the interface provides a clean, well-tested contract for stepping a stateful physical system forward in time, which both the GA's evaluation loop and the separate RL training pipeline (Section 7.2) reuse directly.

Observation space	[altitude, speed, SoC, fuel_kg, P_required_kw]	Action space	[-1, 1] continuous – power-split ratio (PSR)
PSR = 1	100% battery-electric	PSR = 0	100% engine
PSR Engine drives propeller and battery- < 0 charging generator simultaneously			

6.2 Mission-Leg Definition and Sequencing

A mission is specified as an ordered list of **legs**, each either a *cruise* leg (altitude, speed, ground distance, optional signed headwind/tailwind) or a *loiter* leg (altitude, speed, required on-station duration). Legs are validated strictly at input time — invalid combinations (e.g. headwind exceeding airspeed, altitude too close to the base elevation) are rejected outright rather than silently clamped. Between legs, the aircraft automatically flies the fixed-phase segments — takeoff, climb-out, descent, and landing — using fixed speed and climb-rate targets rather than leg-configurable ones:

Phase	Speed target	Climb-rate target
Takeoff	$\max(1.15 \cdot V_{\text{stall}}, 25 \text{ m/s})$	+3.0 m/s (excess-power limited)
Climb	$\max(1.3 \cdot V_{\text{stall}}, 35 \text{ m/s})$	+5.0 m/s (excess-power limited)
Cruise (leg-defined)	$\max(\text{leg speed}, 1.25 \cdot V_{\text{stall}})$	0
Loiter (leg-defined)	$\max(\text{leg speed}, 1.2 \cdot V_{\text{stall}})$	0
Descent	$\max(1.2 \cdot V_{\text{stall}}, 30 \text{ m/s})$	-1.5 m/s (fixed, gravity-driven)
Landing	$\max(1.1 \cdot V_{\text{stall}}, 22 \text{ m/s})$	-0.8 m/s (fixed, gravity-driven)

Cruise legs complete when accumulated ground distance (airspeed corrected for headwind/tailwind) meets the requested distance; loiter legs complete when elapsed on-station time meets the required duration. After the final scheduled leg, the aircraft enters an open-ended "bonus reserve" loiter that continues until fuel or battery reserves become resource-critical, at which point the simulator triggers a return-to-base

sequence (Section 6.4). This bonus loiter is what allows total simulated endurance to exceed the sum of the mission's required legs, and is directly relevant to the redundancy findings in Section 10.6 and Section 11.

6.3 Excess-Power-Limited Climb

Climb rate during Takeoff and Climb phases is not simply commanded; it is limited by the actual excess shaft power available after subtracting level-flight aerodynamic power draw from the maximum power the engine and battery can jointly deliver at the current altitude, fuel state, and state of charge. This prevents the simulator from ever crediting a climb the propulsion system cannot physically sustain — an undersized design correctly climbs more slowly (or not at all) rather than climbing at the commanded rate regardless of available power:

```

P_elec_avail  = 0 if SoC ≤ soc_min else max_battery_power(peak, temp_penalty)
P_engine_avail = 0 if fuel_remaining ≤ 0.01 else max_engine_power(altitude, peak)
P_avail_thrust = (P_elec_avail + P_engine_avail) · η_prop

P_excess  = P_avail_thrust - P_aero          [P_aero at climb_rate = 0]
Vz_actual = max(0, P_excess · 1000 / (W · g))
climb_rate = min(climb_rate_target, Vz_actual)

power_deficit_flag = (Vz_actual < climb_rate_target)
climb_prevented    = (Vz_actual ≤ 0)

```

Eq. 6.1

6.4 Safety and Termination Logic

- **Stall limit:** simulation terminates immediately if the actual lift coefficient exceeds 1.6.
- **Resource-critical return-to-base (RTB):** triggered when $(\text{fuel_ratio} < 0.08 \text{ AND } \text{SoC} < 0.15)$ OR $(\text{fuel_ratio} < 0.03)$ — the current leg is abandoned and logged incomplete, and the aircraft returns to the base elevation and lands.
- **Sustained power deficit:** three consecutive timesteps with a power shortfall greater than 1 kW is treated as a mission-ending condition.
- **Out-of-energy:** both state of charge and fuel simultaneously at their floor values.
- **Safety truncation:** a hard 30-hour (108,000 s) simulated-time limit regardless of remaining resources.

6.5 Reserve and Redundancy Metrics

At the end of a completed episode, the simulator computes STANAG 4671-style single-fault reserve metrics against a canonical reference condition (mean mission-leg altitude, 76% of mean mission-leg speed):

```

Fuel reserve (min) = fuel_remaining / (SFC_eff(ref_P_req) · ref_P_req / 60)

Battery reserve (min) = usable_SoC_above_floor · effective_battery_capacity
                      / ((ref_P_req / η_motor) / 60)

Engine-out survivability, evaluated at the MINIMUM SoC seen anywhere in the flight

```

Eq. 6.2

(not the final SoC):

```
survivable_min = usable_SoC_at_min · effective_battery_capacity  
                / ((ref_P_req /  $\eta_{\text{motor}}$ ) / 60)  
engine_out_survivable = (survivable_min ≥ 30.0)    [STANAG 4671 30-minute requirement]
```

Evaluating survivability at the minimum SoC seen at any point in the flight — rather than only at mission end — is a deliberately conservative choice, and its consequences are discussed in Section 10.6.

7. Energy Management Strategy

7.1 Rule-Based Heuristic Policy

The default power-split policy is a phase-and-state-machine heuristic (in the tradition of rule-based hybrid-electric energy management strategies described in the literature), prioritizing electric power during low-speed, high-torque phases and reserving the engine for sustained, high-load cruise where its specific fuel consumption is most favorable:

Phase	Power-split ratio (fraction from motor)
Takeoff	0.20 if SoC < 0.3, else 0.45
Climb	0.0 if SoC < 0.4; 0.10 if SoC < 0.6; else 0.15 (hard-capped at 0.05 as a battery-preservation guard)
Cruise	0.10 if SoC > 0.85; 0.0 if SoC < 0.3; else 0.05
Loiter — Silent Mode (electric-only)	1.0 — engaged when silent-loiter is requested and SoC > 0.15
Loiter — low-fuel override	0.60 — engaged when fuel ratio < 0.15 and SoC > 0.12
Loiter — Endurance Mode (engine-only)	0.0 — used once battery is too depleted for stealth operation
Descent / Landing	0.0 (gravity-assisted, no shaft power split needed)

Switching out of Silent Loiter mode back to the engine incurs a one-time 0.05 kg fuel spike and forces the first restart step's engine output to 50% of commanded power, modelling the non-instantaneous, non-free relight of a real turboshaft rather than an idealized instant engine start.

7.2 Reinforcement-Learning Policy

As an alternative to the rule-based heuristic, a Proximal Policy Optimization (PPO) policy is trained using Stable-Baselines3 on a purpose-built training environment that generates randomized single-leg scenarios (cruise or loiter, altitude 2,500–6,000 m) with a 60% probability of an injected scripted disturbance — a temperature drop (–30 to –5 °C), elevated turbulence (0.5–1.0), and/or a wind gust (+20 to +60 km/h) triggered at a randomized point in the leg. The policy observes a 9-dimensional feature vector (altitude, speed, SoC, fuel ratio, required power, density ratio, temperature, turbulence level, and disturbance-active flag) and outputs a power-split ratio clipped to [0, 1].

The trained policy is evaluated against the rule-based heuristic under an identical scripted disturbance (a climb-phase shock — the phase with the least power margin — combining –35 °C, maximum turbulence, and a +90 km/h gust) using a fixed engine/battery configuration, with both endurance and power-deficit outcomes reported directly rather than assumed. The API exposes this as a selectable `policy_mode`

(`heuristic` or `r1`) applied only at the re-simulation stage; the GA sizing loop itself always sizes against the heuristic policy for determinism and speed.

8. Optimization Methodology

8.1 Genetic Algorithm Formulation

The outer sizing loop is a real-valued genetic algorithm implemented with DEAP. Each individual encodes a candidate propulsion design as a fixed-length vector of genes; each generation applies selection, crossover, and mutation to evolve the population toward higher-fitness (longer-endurance) designs, while a Hall of Fame retains the single best individual seen across the entire run.

8.2 Search Space and Hyperparameters

Design variable	Range / domain	GA hyperparameter	Value
Engine size (kW)	[30.0, 120.0], continuous	Population size	40
Battery capacity (kWh)	[16.69, 50.0], continuous	Generations	15
Motor count	{1, 2, 4}, rounded from a continuous gene	Crossover	Blend ($\alpha = 0.5$), $P = 0.6$
Cruise PSR (optional)	[0.0, 1.0]	Mutation	Gaussian, $\sigma = [8.0, 4.0, 0.5]$ (+ [0.15, 0.15] if PSR-searched), $P = 0.35$, per-gene $P = 0.35$
Loiter PSR (optional)	[0.0, 1.0]	Selection	Tournament, size 3

Bounds are hard-clipped after every crossover and mutation operation, so the search never evaluates a physically undefined design outside the configured ranges. Convergence statistics (max, min, and average population fitness per generation) are captured for post-run analysis (Section 10.2).

8.3 Fitness Function

Fitness is the raw mission endurance in hours as reported by the full physics simulation, with a single multiplicative penalty applied once — not stacked — if the mission did not conclude in a safe landing or if any required loiter leg's on-station duration was not fully met:

```
fitness = time_elapsed_hours
Eq. 8.1

if NOT landed_safely OR ANY required loiter leg incomplete:
    fitness *= 0.4    [INCOMPLETE_MISSION_PENALTY]

if fuel_initial ≤ 0 (design exceeds MTOW budget) OR environment construction raises an
```

```
error:
```

```
    fitness = 0.0      (individual rejected without simulation)
```

This formulation rewards designs that both fly long *and* complete the mission as specified — a design that merely maximizes raw hours by, for example, failing to complete a required loiter and running out of fuel mid-flight is penalized rather than rewarded for accidental longevity.

9. Software Implementation

9.1 Backend REST API

The backend exposes a FastAPI application (AeroOptima – Hybrid-Electric UAV Propulsion Optimization API , version 3.0.0) with request/response models defined as Pydantic v2 schemas, including field-level validation (e.g. altitude bounds, headwind-versus-speed consistency) that rejects malformed mission specifications before any simulation work begins.

Endpoint	Method	Description
/api/optimize	POST	Accepts a full mission specification (legs, base elevation, payload, fuel fraction, ambient temperature, turbulence, silent-loiter flag, battery chemistry, policy mode, optional environmental disturbance). Runs the GA sizing loop, re-simulates the winning design at fine time resolution for clean telemetry, runs the sensitivity sweep, and returns the optimal specification, full telemetry trace, and GA convergence statistics.
/api/health	GET	Liveness check; returns service status and the configured motor model.

The optimization response includes the sized engine/battery/motor configuration, weight breakdown, endurance (with a sensitivity-derived min/max band), per-leg completion status, reserve-equivalent minutes for both fuel and battery, the engine-out survivability flag, and a full per-timestep telemetry array (altitude, speed, power required/delivered/motor/engine, state of charge, fuel remaining, weight, flight phase, power-split ratio, effective SFC, and disturbance-active flag) for downstream visualization.

9.2 Frontend Dashboard

The frontend is a single-page Next.js dashboard providing an ordered mission-leg builder, environmental and advanced-option controls (base elevation, ambient temperature, turbulence, payload, initial fuel fraction, silent-loiter toggle, battery chemistry, policy mode, and a scripted-disturbance panel), and two quick-load demonstration presets covering a high-altitude cold-weather stealth loiter and a storm-front contingency mission. Results are rendered as a status summary (endurance, on-station status, sized specifications), a mission-phase timeline, an MTOW weight-budget stacked bar, a system-constants panel, and a tabbed 3D flight-path replay / telemetry-chart view with playback scrubbing that highlights disturbance-active time windows.

9.3 Testing and Regression Strategy

Correctness is protected at two levels. Physics smoke tests validate individual mechanisms in isolation — altitude derating monotonicity, correct prevention of an unphysical "phantom" climb when a design is genuinely underpowered, correct handling of same-altitude consecutive legs, and byte-for-byte reproducibility when no environmental disturbance is specified. A separate, deterministic (seed-pinned) regression suite compares full simulation output — endurance, fuel burned, complete state-of-charge and

fuel traces, and termination reason — against a golden-baseline result set to catch any unintended behavioral change introduced by future modifications to the physics core.

10. Results and Validation

10.1 Canonical Sizing Result

Against the platform's documented baseline mission (a long-range cruise-loiter-cruise profile), the GA converges on an engine/battery/motor combination close to the battery search floor, reflecting that once the engine is large enough to sustain cruise with margin, additional battery capacity beyond the airworthiness-derived minimum trades away fuel-carrying weight without a compensating endurance benefit for this mission shape:

Engine rating	≈ 64.36 kW	Battery capacity ≈ 16.69 kWh (search-space floor)
Motor count	1	Nominal endurance (engine-sustained cruise) ≈ 21.0 h
Silent-loiter endurance (battery-only)	≈ 34 min	Minimum SoC observed in flight 0.255 (floor: 0.10)

Single-engine-failure survivability for this configuration is satisfied for the required mission legs: the sized engine's continuous rating exceeds the cruise power requirement with margin, so a battery failure does not prevent completing the mission on engine power alone; and the battery, sized to the 30-minute STANAG-derived floor, sustains the mandated single-fault electric-only reserve within the scope of the required legs (see Section 10.6 for the important caveat regarding the open-ended bonus loiter extension).

10.2 Genetic Algorithm Convergence

A representative optimization run (population 40, generations 15) illustrates typical convergence behavior: fitness rises sharply once the population discovers a first "landing-safe, mission-complete" design region around generation 5–8 (the fitness landscape has a hard discontinuity here — the ×0.4 incomplete-mission penalty of Section 8.3 — so early generations cluster near zero until that region is found), then improves more gradually as the engine/battery combination is fine-tuned:

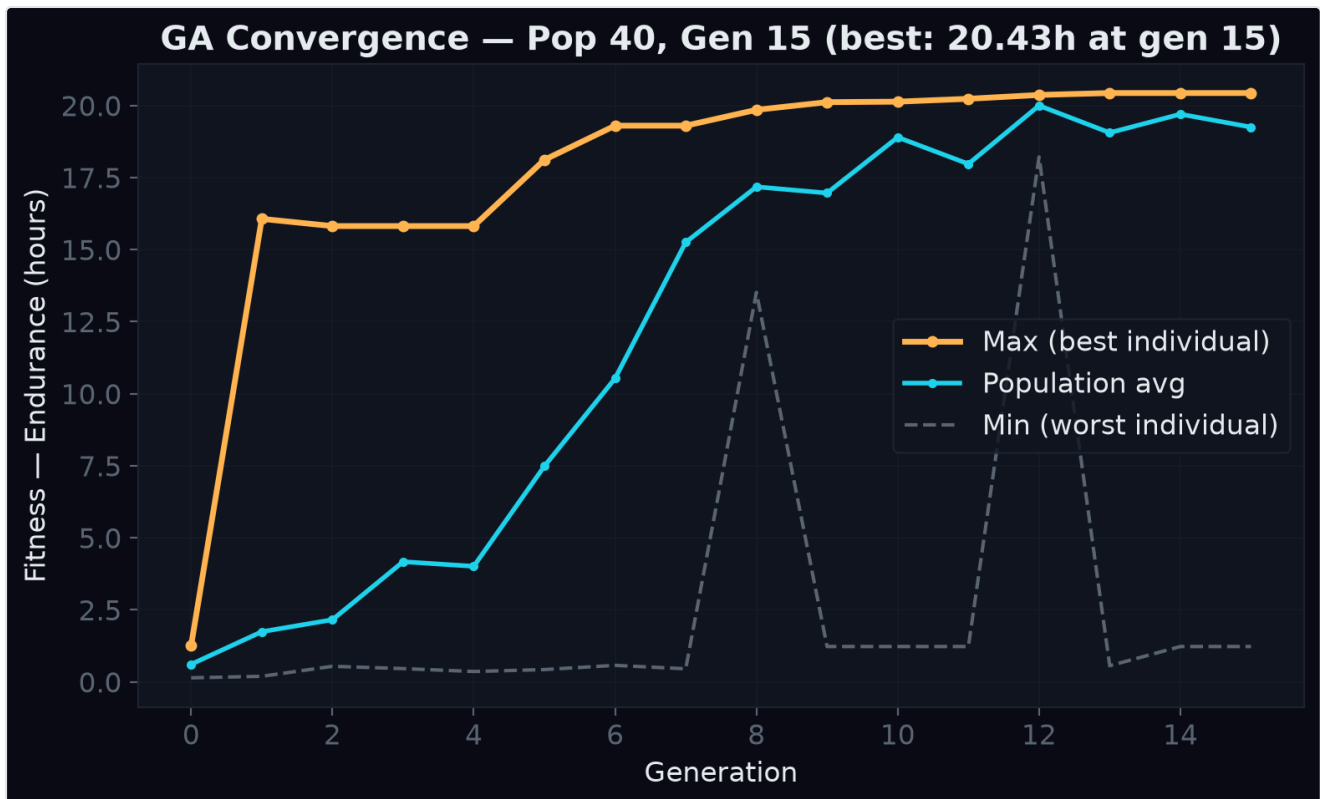


Figure 10.1 — Best, average, and worst population fitness by generation for a representative sizing run (best result: 20.43 h at engine ≈ 115.0 kW, battery 16.69 kWh, 1 motor).

The discontinuous jump visible between generations 7 and 8 reflects tournament selection rapidly propagating a single newly-discovered mission-complete individual through the population once it exists — evidence that the search is being driven by the fitness discontinuity described in Section 8.3 rather than a smooth gradient, which is expected given how the penalty is structured.

10.3 Sensitivity Analysis

To characterize how sensitive the sized design's endurance is to component-level uncertainty, the platform automatically re-simulates the GA-selected design under four combinations of $\pm 10\%$ specific fuel consumption and $\pm 10\%$ battery energy density relative to the nominal data-sheet values:

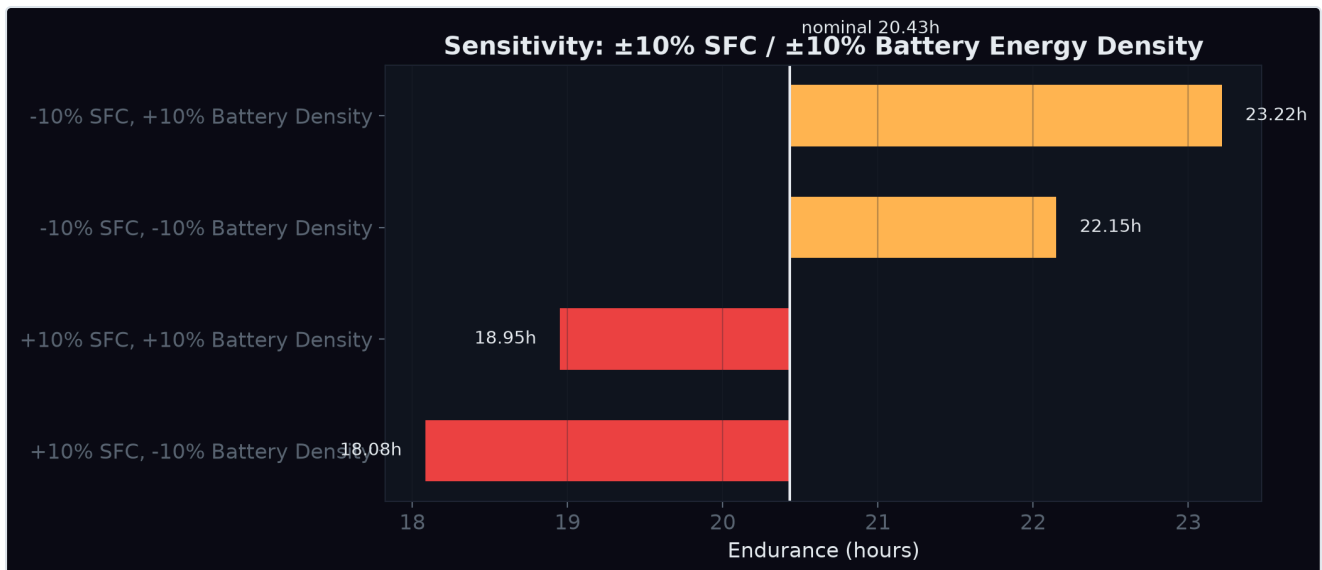


Figure 10.2 — Endurance sensitivity to $\pm 10\%$ SFC and $\pm 10\%$ battery energy density around a 20.43 h nominal design point.
Range: 18.08 h – 23.22 h ($\approx -11.5\%$ / $+13.6\%$).

Fuel consumption dominates the sensitivity band, which is consistent with the design being an engine-cruise-dominated mission: a 10% SFC penalty costs more endurance than a 10% battery energy-density penalty recovers, and vice versa. This sweep is returned automatically with every optimization response as a min/max endurance band, giving a first-order robustness estimate without requiring a separate Monte Carlo run.

10.4 Weight Budget Breakdown

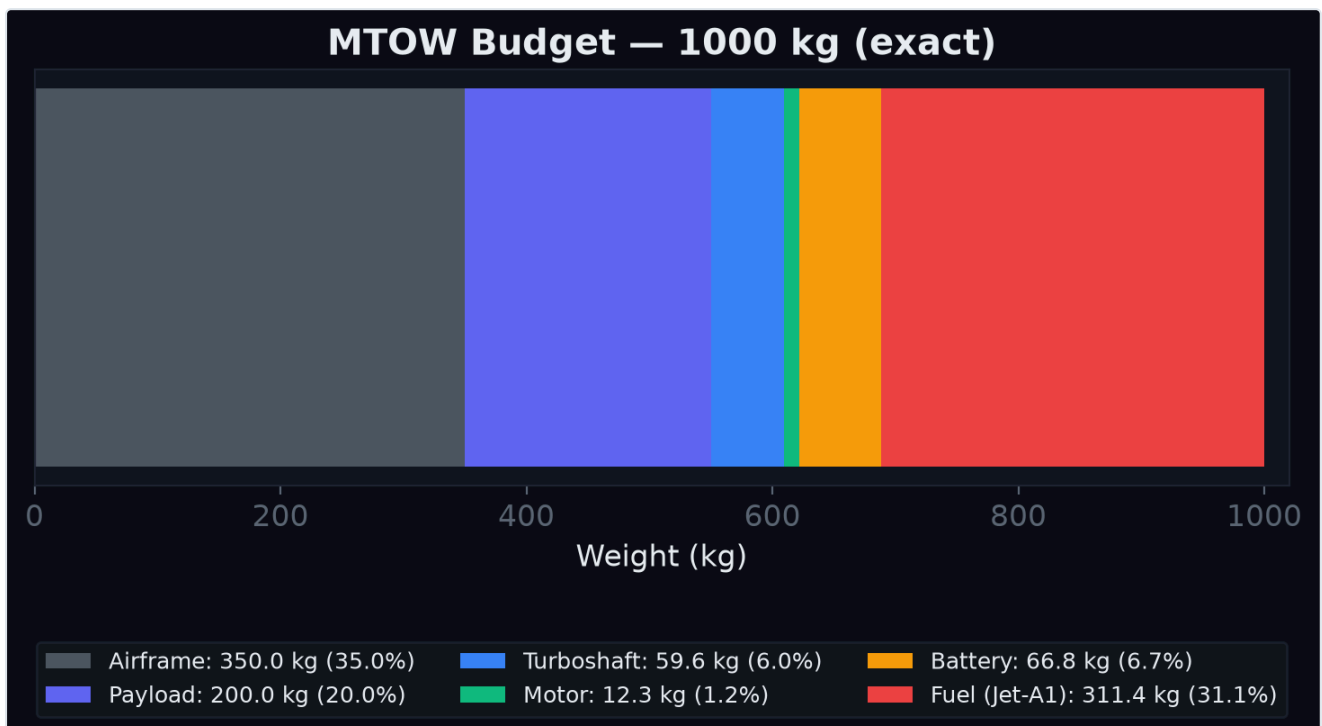


Figure 10.3 — MTOW budget for a representative sized design, closing exactly to 1,000 kg.

Across every design point examined in this report and in the multi-scenario validation (Section 10.6), the weight budget closes exactly to the 1,000 kg MTOW constraint by construction — fuel mass is always the residual after airframe, payload, engine, motor, and battery weight are subtracted, so the constraint cannot be silently violated by the optimizer.

10.5 Comparative Positioning

Positioning the sized design against publicly documented specifications of fielded fixed-wing UAV platforms of broadly comparable weight class gives a sanity check on whether the endurance figures the tool produces are in a physically credible range, rather than an artifact of the model:

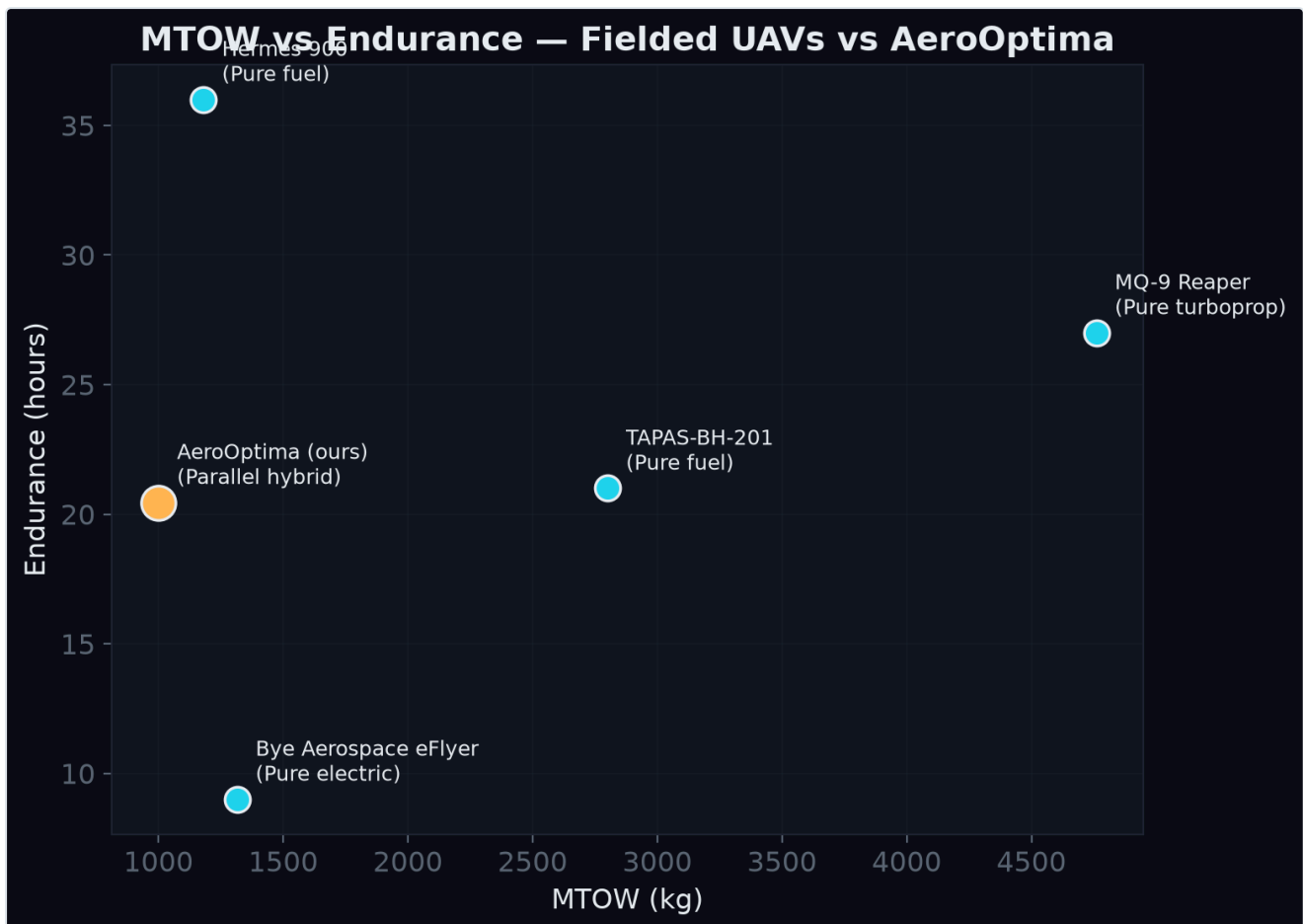


Figure 10.4 — MTOW vs. endurance for several fielded platforms (published specifications) against the AeroOptima-sized parallel-hybrid design.

The AeroOptima design sits between pure-electric platforms (short endurance, no engine weight penalty) and pure-fuel/turboprop platforms of considerably higher MTOW, which is the expected position for a parallel-hybrid architecture at this weight class: it does not out-endure a purpose-built heavy fuel-only platform, but it substantially exceeds pure-electric endurance at a comparable or lower MTOW, which is the value proposition the hybrid architecture is intended to deliver.

10.6 Multi-Scenario Validation

To test whether the sizing tool generalizes beyond a single bundled mission, six scenarios spanning distinct operating theatres were constructed and run end-to-end through the unmodified GA and simulator: a high-altitude cold-weather endurance loiter, the same mission under continuous stealth loiter as a stress test, a hot-desert profile, a temperate-plains profile, a monsoon-belt profile with moderate turbulence, and a maritime coastal-patrol profile.

Scenario	Base alt.	Ambient temp.	Sized engine/battery/motors	Outcome	Engine-out survivable
High-altitude, cold, endurance loiter	4,130 m	−10 °C	108.05 kW / 16.69 kWh / 1	Completed	No
Same, continuous stealth loiter (stress test)	4,130 m	−10 °C	59.28 kW / 46.84 kWh / 2	Power deficit	No
Hot desert	200 m	45 °C	65.61 kW / 18.16 kWh / 1	RTB failure (sampled run)	No
Temperate plains	250 m	25 °C	55.60 kW / 16.69 kWh / 1	Completed	No
Monsoon-belt, moderate turbulence	3,000 m	5 °C	89.72 kW / 16.69 kWh / 1	RTB failure (sampled run)	No
Maritime coastal patrol	0 m	32 °C	94.57 kW / 16.69 kWh / 1	Completed	No

What the validation confirms

The weight budget closes to exactly 1,000 kg in every scenario regardless of theatre. Engine sizing is theatre-sensitive rather than a fixed answer — the calmest, lowest-altitude, shortest-transit scenario (temperate plains) sized the smallest engine (55.6 kW), while high-altitude and long-transit scenarios sized substantially larger engines (90–108 kW), which is the physically correct direction of adaptation. The battery repeatedly converges to, or near, the airworthiness-derived 16.69 kWh floor across most scenarios, confirming that finding generalizes rather than being specific to one mission.

What the validation exposes as current limitations

Engine-out survivability came back false in every scenario that otherwise completed successfully. The 30-minute single-fault reserve requirement is evaluated at the minimum state of charge observed anywhere in the flight (Section 6.5), and in practice that minimum is typically reached deep into the open-ended bonus-loiter extension (Section 6.2) rather than during the mandatory legs — so the reserve requirement holds for the mission as specified, but not for the maximum-endurance extension the heuristic policy continues to fly once the required legs are satisfied. Two of the three scenarios run under non-zero turbulence failed to safely return to base on the specific random draw tested, both after the on-station requirement had already been met, on the return-transit leg. Continuous, multi-hour, fully electric stealth loiter is infeasible at any design point

reachable within the current search space, at either a 6-hour or a 12-hour duration. These findings, and their engineering implications, are carried forward into Section 11.

11. Limitations and Future Work

11.1 Engine-out survivability under the bonus-loiter extension

The current single-fault reserve claim is accurate for the mission as explicitly specified by its required legs, but does not hold once the aircraft is flown to the fuel/battery-exhaustion point the simulator's open-ended bonus loiter allows. The most direct fix is to bound how long the bonus-loiter extension is permitted to run before triggering RTB, so that the aircraft is never flown into a state from which it could not survive a single-point propulsion failure; a secondary or complementary fix is to report survivability as an explicit function of "required mission" versus "maximum-endurance extension" rather than a single boolean.

11.2 Return-to-base robustness under turbulence

Because turbulence is sampled stochastically per timestep, a single re-simulation of a GA-selected design is one sample from a distribution of possible outcomes, not a guarantee. The observed RTB failures under non-zero turbulence indicate that a design should be evaluated across multiple random seeds — and ideally optimized for a worst-case or percentile outcome rather than a single nominal run — before its robustness under realistic turbulent conditions is asserted.

11.3 One-sided battery thermal model

The battery derating model penalizes capacity and power only below +10 °C; no corresponding derating exists above the point at which real lithium-ion packs also lose usable capacity and safe discharge rate. This makes hot-climate scenarios more optimistic in simulation than a real deployed pack would likely achieve, and is a direct, scoped candidate for the next modelling iteration.

11.4 Continuous full-electric stealth loiter

At the airframe and component weight classes modelled, continuous multi-hour fully-electric loiter is not achievable at any design point the current search space can reach. Extending feasibility would require either a battery energy-density assumption beyond the current chemistry presets, a reduced-power stealth-loiter speed/altitude profile, or accepting a shorter continuous silent-loiter duration as the realistic operational envelope, and stating that duration explicitly rather than treating unlimited silent loiter as an available mode.

11.5 Discrete-timestep floor overshoot

Minimum state-of-charge is occasionally observed slightly below the nominal SoC floor (for example, 0.087 against a 0.10 floor) due to a 60-second simulation timestep crossing the floor within a single step before being clamped. This is a minor numerical artifact rather than a modelling error, but is worth tightening (e.g. via adaptive sub-stepping near the floor) if downstream reserve calculations are made more sensitive to it.

11.6 Sizing determinism versus policy choice

The GA sizing loop always evaluates candidates using the rule-based heuristic policy for speed and reproducibility, even when the eventual re-simulation is requested under the learned RL policy. This means the RL policy is currently evaluated against, but does not influence, the sizing decision itself. A natural extension is a sizing mode that evaluates fitness under the RL policy directly, or under a policy-agnostic

worst-case power-availability check, so that sizing and the eventual operating policy are consistent by construction.

12. Conclusion

AeroOptima demonstrates that hybrid-electric propulsion sizing for a fixed-wing tactical UAV can be treated as a direct output of coupling an evolutionary search with a physically detailed, mission-aware flight simulation, rather than as a static reference design applied uniformly across missions. The platform's engine and battery sizing responds correctly to changes in altitude, temperature, mission distance, and loiter duration; its weight budget is closed and constraint-respecting by construction; and its energy-management layer offers both an interpretable rule-based baseline and a learned alternative evaluated under realistic environmental disturbance.

The independent multi-scenario validation exercise (Section 10.6) is presented in this report not as a demonstration of unconditional readiness, but as evidence that the platform's own outputs are the right tool for identifying exactly where its current physical and control assumptions break down — the bonus-loiter reserve gap, turbulence-driven RTB fragility, and the one-sided thermal model are concrete, scoped engineering priorities rather than open-ended uncertainty. That combination — a sizing tool whose own validation surfaces its real limitations rather than only its favorable results — is the most defensible basis for further hardening the model ahead of any operational use.

A. Appendix A — Glossary of Terms

PSR

Power Split Ratio — fraction of required shaft power drawn from the electric motor (0 = pure engine, 1 = pure electric; negative values indicate the engine simultaneously charging the battery via a generator).

SFC

Specific Fuel Consumption (kg/kWh) — fuel mass burned per unit of mechanical energy produced by the engine.

SoC

State of Charge — battery charge level, normalized to the range [0, 1].

MTOW

Maximum Takeoff Weight — the hard 1,000 kg upper bound on total aircraft weight.

ISA

International Standard Atmosphere — the standardized temperature/density-versus-altitude reference model.

AR

Aspect Ratio = $\text{wingspan}^2 / \text{wing area}$ = 16.07 for this airframe; higher aspect ratio reduces induced drag.

CL / CD

Lift coefficient / drag coefficient.

C-rate

Battery discharge rate expressed as a multiple of pack capacity (e.g. 3C on a 20 kWh pack corresponds to a 60 kW draw).

AMSL

Above Mean Sea Level — the altitude reference frame used for all base and leg altitudes.

DEAP

Distributed Evolutionary Algorithms in Python — the genetic-algorithm framework used for the sizing loop.

GA

Genetic Algorithm.

RTB

Return to Base.

Gagg-Ferrar

A 1934 empirical model for naturally-aspirated engine power loss with altitude, used here to derate the turboshaft.

Prandtl-Glauert

A compressibility correction to drag as airspeed approaches the transonic regime, applied here between Mach 0.4 and 0.95.

PPO

Proximal Policy Optimization — the reinforcement-learning algorithm used to train the alternative power-management policy.

STANAG 4671

A NATO standardization agreement covering airworthiness requirements for unmanned aircraft systems, referenced here specifically for its 30-minute single-propulsion-fault reserve requirement.

B. Appendix B — Formula Reference Summary

#	Formula	Purpose
4.1	$\rho(h)$, $T(h)$ with configurable T_0	Atmosphere / density-altitude model
4.2	$a(T)$, M , V_{stall}	Speed of sound, Mach number, stall speed
4.3	CL , CD , P_{aero} , P_{climb} , P_{shaft}	Oswald drag polar + compressibility, power required
4.4	J , η_{prop}	Advance-ratio propeller efficiency
5.1	weight_engine , $\text{engine_continuous/peak_kw}$	Engine weight and rating scaling
5.2	σ , derate , $P_{\text{max_engine}}(h)$	Gagg–Ferrar altitude derating
5.3	$\text{SFC}(\text{load_fraction})$, fuel_burned	Partial-load fuel-consumption penalty
5.4	weight_battery , $P_{\text{max_battery}}$, temp_penalty	Battery weight, power limit, cold derating
5.5	$\text{weight_empty_and_payload}$, fuel_initial	Weight budget / MTOW closure
6.1	P_{excess} , V_{z_actual} , climb_rate	Excess-power-limited climb ("phantom climb" fix)
6.2	Fuel/battery reserve minutes, $\text{engine_out_survivable}$	STANAG-style reserve and redundancy metrics
8.1	$\text{fitness} = \text{time_elapsed_hours} \times \text{penalty}$	Genetic-algorithm fitness function